

Financial Cascade

Frontend & UX Architecture Guide — Portfolio Data Ingestion Strategy

KEY ARCHITECTURAL QUESTION

"How are we getting users' financial portfolio data? Do we connect to bank accounts like budgeting apps (Plaid API), or is it manual input?"

Core Ingestion Strategy

Third-party bank aggregators (e.g., Plaid) require paid credentials, corporate security audits, and complex OAuth handshakes that are impractical for an academic scope. Instead, **Financial Cascade** utilizes three intuitive, friction-free frontend data entry mechanisms:

1. Interactive Canvas Input

Users visually construct their portfolio directly on the D3.js canvas:

- **Add Node Modal:** Select type (Income, Checking, Savings, Credit Card, 401k/Roth IRA).
- **Node Parameters:** Input Balance, APR/Yield %, and Monthly Flow.
- **Visual Edges:** Drag connecting lines representing cash flow targets (e.g., \$3,000/mo from Salary to Checking).

2. Instant Demo Archetypes

A dropdown menu instantly populates the canvas with pre-built profiles for rapid testing and presentations:

- **"Recent Rowan Grad":** \$65k Salary, \$25k Student Debt @ 5.5%, \$2k Checking.
- **"Mid-Career":** \$110k Salary, \$350k Mortgage @ 6.2%, \$60k 401(k).
- **"Aggressive Payoff":** Multiple high-APR credit cards to demo AI optimization.

3. CSV / JSON Import Support (Optional Batch Entry)

Users can upload a simple CSV file to generate complex node networks in milliseconds:

```
NodeName,Type,Balance,APR,FlowTarget,FlowAmount Salary,INCOME,75000,0.0,Checking,4500
Checking,ACCOUNT,3000,0.05,StudentLoan,400 StudentLoan,DEBT,18000,6.5,None,0
```

Why This Strategy Fits Our Project & Evaluation Criteria

- **Privacy & Security:** No credentials, SSNs, or sensitive banking tokens are required or stored.
- **Scenario Planning ("What-If" Analysis):** Live bank links only reflect past state. Manual & archetype inputs allow users to test hypothetical scenarios (e.g., "What if I get a \$10k raise?").
- **Zero Presentation Friction:** Evaluators and professors can run full interactive demos in seconds without needing a real bank account setup.

